Technical Specification Document: AI-Driven Plagiarism Intelligence for Assignments

1. Problem Statement Overview

- **Problem Statement No:** 10
- **Title:** AI-Driven Plagiarism Intelligence for Assignments
- **Target Audience:** Academic Institutions, Evaluators, and Instructors
- **Mandatory Technology Requirement:** IBM Cloud / IBM watsonx.ai / IBM Granite Models

2. The Core Challenge

Academic institutions face increasing difficulty in detecting nuanced forms of academic misconduct, particularly when assignment submissions are heavily paraphrased, reordered, or generated autonomously by LLM applications. Traditional plagiarism tools rely on rigid string-matching algorithms and completely lack contextual sensitivity to an individual student's historical writing footprint or baseline style.

Without an automated, context-aware method to evaluate individual linguistic patterns and track historic grader feedback loops, evaluators cannot reliably flag artificial text or structural anomalies in student assignments.

3. The Objective & Solution Scope

The objective is to build an AI-powered, multi-agent Plagiarism Intelligence system that delivers context-aware integrity verification. The system moves away from global database scraping and focuses on localized student history vectors to:

- **Establish a Linguistic Persona Baseline:** Map unique writing footprints based on vocabulary complexity and sentence structures.
- **Execute Contextual RAG Retrieval:** Dynamically extract historical student submissions and past instructor comments from a local vector repository before processing.
- **Process Multi-Modal Uploads:** Integrate a local OCR engine to read text from both digital uploads and images of printed/handwritten assignments.
- **Detect Structural Anomalies:** Pinpoint unnatural vocabulary inflation or patterns typical of generative AI tools.

4. Architectural Stack

- **Frontend Dashboard:** React.js (TypeScript) + Vite
- **Backend Application Server:** FastAPI (Python)
- **Image Processing Engine:** Local Pytesseract OCR
- **AI Core Infrastructure:** IBM watsonx.ai Platform